DCFMS Gateway: Context Compression and Data Exposure Reduction for Large Language Model Workflows

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Abstract

DCFMS Gateway is an experimental context-compression layer for large language model workflows. It reduces the amount of text sent to the model by selecting and compressing a smaller, task-relevant context from a larger corpus. The intended effect is lower token usage, lower cost, and reduced information exposure compared with sending full documents or complete source collections to the model.

This paper summarizes internal public-release evidence for the system using benchmark summaries already present in the repository. The evidence includes a baseline OFF/ON comparison, a local token-compression benchmark, client-demo runs, and a laboratory ordered-context benchmark. These results are internal measurements only and have not yet undergone independent third-party validation.

The goal is not only token reduction. The key objective is preserving answer quality while reducing context size. The public evidence therefore evaluates token reduction and answer quality together, not as separate success criteria. Compression is useful only when the resulting answer remains grounded, accurate, and fit for purpose.

Contents

Executive Summary

DCFMS Gateway is a private context-reduction layer for LLM workflows. It sits before the model, keeps the full corpus on the user side, and forwards only selected and compressed context to the external model. That makes it relevant not just for token reduction, but also for cost control and for limiting how much source material is exposed to the model.

The public release snapshot shows three consistent signals:

- token usage can be reduced materially,
- answer quality can remain stable or improve after corpus refinement,
- internal benchmark evidence is promising but still awaits independent validation.

The goal is not only token reduction. The key objective is preserving answer quality while reducing context size. The release package therefore reports quality counts alongside compression metrics and treats ROI as an estimate, not a billing statement.

1 Key Results

Metric	Result
Compression Ratio	up to 119.93x
Noise Removed	up to 98.61%
Full20 OFF/ON	20/0/0 → 20/0/0
Client Demo Improvement	17/0/3 → 18/1/1
Ordered Context Improvement	3/2/15 → 10/1/9
Data Exposure	Reduced via context minimization
Validation Status	Internal benchmarks only

Table 1: Key benchmark signals from the public release snapshot.

These results summarize the strongest signals observed across the public benchmark set. They should be interpreted as internal benchmark evidence rather than independently validated results.

2 Introduction

Large language model workflows often fail in predictable ways when they are fed too much context. Long prompts cost more, introduce more noise, and make it harder to control what the model actually uses. In retrieval-heavy workflows, the issue is often not a lack of information but an excess of it.

DCFMS Gateway addresses this by reducing the context before it reaches the model. The system is positioned as a practical layer for document-heavy or corpus-heavy workflows where context size, answer quality, and data exposure all matter at the same time.

This paper is intentionally conservative. It reports internal benchmark evidence, avoids implementation detail, and does not claim independent validation.

3 Problem Statement

The problem addressed by DCFMS Gateway has four parts.

- Full context is expensive because every extra token contributes to model cost.
- Full context can also be noise, because large corpora often contain material that is irrelevant to the question.
- Full context increases data exposure, because more source material is transmitted to the model than is strictly needed.
- Classical prompting and many retrieval pipelines still send too much information upstream.

The practical question is therefore not whether a model can read everything. The practical question is whether the model can answer well with less context.

4 DCFMS Gateway Overview

DCFMS Gateway is a private reduction layer that operates before the LLM call. Its high-level behavior is simple:

- the full corpus remains in the user-controlled environment,
- only selected and compressed context is forwarded to the model,
- the exposed context is smaller than the original corpus,
- the system is evaluated as a context-control layer, not as a new model.

This document does not describe implementation internals, retrieval mechanics, or engine-level details. The public release scope is limited to observable behavior and benchmark outcomes.

5 Methodology

The release package compares raw/full context against compressed context using the same broad measurement approach across benchmark families.

5.1 Core Metrics

The main metrics used in the public material are:

Metric	Type	Meaning
tokens_before	measured	Token count before compression or before reduction.
tokens_after	measured	Token count after compression.
compression_ratio	measured	Ratio of raw tokens to compressed tokens. Higher is better for reduction.
noise removed %	measured	Estimated fraction of irrelevant or redundant context removed by compression.
good / partial / poor	measured	Quality counts describing whether the answer stayed grounded and useful.
ROI values	estimated or projected	Scenario-based cost impact derived from token assumptions and token pricing.

5.2 Why good / partial / poor

The quality scale is intentionally simple because it is meant to reflect answer usefulness rather than only retrieval statistics.

Label	Meaning	Why it matters
good	grounded, relevant, and materially correct	Indicates that compression preserved answer quality.
partial	partly grounded or directionally correct, but incomplete or generic	Captures answers that are usable but not ideal.
poor	incorrect, ungrounded, or outside the intended corpus	Flags cases where compression or corpus quality was not sufficient.

This scale was used because token reduction alone is not enough. A shorter prompt is only valuable if the answer remains reliable.

5.3 Measured, Estimated, Projected

The public package separates results into three evidence classes:

- **Measured:** benchmark counts, compression ratios, and noise removal values.
- **Estimated:** scenario-based monthly and yearly cost savings.
- **Projected:** larger-scale business-impact scenarios derived from the same assumptions.

6 Benchmark Results

This section summarizes the benchmark evidence used in the release package. The figures shown here are internal benchmark measurements only and should not be interpreted as independently validated results.

6.1 OpenAI Full20 OFF/ON

Benchmark	Quality counts	Interpretation
OpenAI Full20 OFF/ON	20/0/0 -> 20/0/0	Compression was achieved without measurable degradation in answer quality.

Table 4: Baseline comparison between the raw path and the DCFMS path on the Full20 benchmark.

6.2 Local ROI Token Compression

Benchmark	Token result	Interpretation
Local ROI Token Compression	72,602.3 avg -> 5,037.05 avg; 18.94x; 93.06% noise removed	Compression was achieved without measurable degradation in answer quality.

Table 5: Measured token compression on the project-native ROI benchmark.

6.3 Client Demo V1

Benchmark	Result	Interpretation
First client benchmark	20 questions; 6/11/3; 88.7129x; 95.35% noise removed	Corpus refinement was still needed.
API ask smoke after patch	8 questions; 8/0/0; 89.3274x; 98.39% noise removed	Answer quality improved after corpus refinement.
Full20 after patch	20 questions; 18/1/1; 119.9268x; 98.61% noise removed	Answer quality improved after corpus refinement.
Final patch plateau	20 questions; 17/0/3; 118.6608x; 98.5946% noise removed	Compression stayed strong, but quality gaps remained.

Table 6: Client-demo evidence showing that corpus refinement can improve answer quality while preserving high compression.

6.4 Zero Drift Ordered-Context Laboratory Benchmark

Benchmark	Result	Interpretation
Zero Drift ordered-context	20 questions; 3/2/15 -> 10/1/9	Quality improved after corpus refinement and ordering.

Table 7: Laboratory benchmark used as a controlled signal for ordered-context behavior.

6.5 Answer Quality Preservation

The release package emphasizes answer quality preservation explicitly. The following benchmark transitions are included in the source materials:

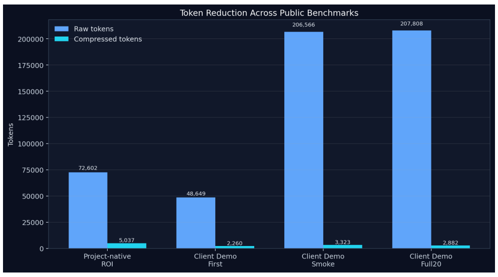
Benchmark	Transition	Note
Full20 OFF/ON	20/0/0 -> 20/0/0	No measurable regression in answer quality.
Client Demo	17/0/3 -> 18/1/1	Quality improved after corpus refinement.
Ordered Context	3/2/15 -> 10/1/9	Quality improved after corpus refinement and ordering.

Table 8: Answer quality preservation examples retained from the public-release materials.

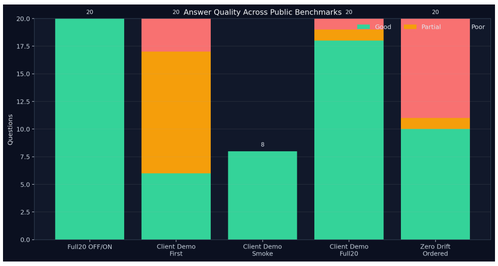
DCFMS is not a compression system for compression's sake. Its objective is reducing context while preserving or improving answer quality.

7 Figures

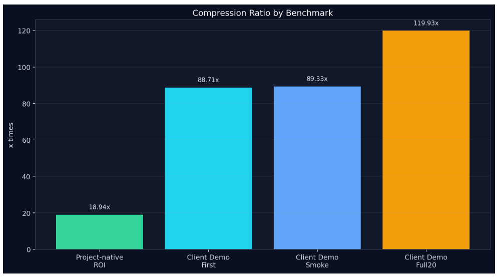
The release package includes four public PNG figures. They are used here to show token reduction and answer quality side by side, and to summarize compression ratio and ROI signals.



(a) Token reduction



(b) Answer quality



(c) Compression ratio

(d) ROI

Figure 1: Public benchmark figures showing token reduction, answer quality, compression ratio, and ROI estimates included in the Zenodo-ready release package.

8 Token Reduction and Cost Impact

Token reduction is a measurable output of DCFMS Gateway, but the public release treats cost impact carefully.

The reported ROI values are scenario-based estimates derived from token usage assumptions. They are useful for reasoning about possible savings, but they are not a live billing audit.

Scenario	Queries	Raw cost	DCFMS cost	Monthly savings
Small business	10,000	$2,000.00	$16.85	$1,983.15
Medium business	50,000	$10,000.00	$84.25	$9,915.75
Enterprise	100,000	$20,000.00	$168.50	$19,831.50

Table 9: ROI scenario estimates from the public release package.

The economic message is conservative: lower token usage can mean lower cost, but the release treats those figures as projections based on the benchmark model.

9 Data Exposure Reduction

Data exposure reduction is a core part of the public narrative, but the wording must remain precise.

DCFMS Gateway reduces exposure because the complete corpus does not need to be transmitted to the model. Only selected and compressed context is sent. Compared with forwarding full documents or entire source collections, this lowers the amount of information exposed to an external LLM.

That statement is intentionally narrower than a security or compliance claim. The release does *not* claim that the system is GDPR compliant, secure, privacy preserving, or certified. It only states the observable behavior: less context leaves the user-controlled environment.

10 Limitations and Threats to Validity

The public release has important limitations.

- The benchmark results are internal.
- No independent third-party validation has been completed yet.
- The corpus set is limited and may not generalize to all domains.
- ROI is estimated, not taken from live billing records.
- Broad comparison against Haystack, LlamaIndex, and classical RAG is not yet included.
- Results may depend strongly on corpus quality and on the question set.
- Laboratory benchmarks are informative, but they are not the same as production customer evidence.

These limitations do not invalidate the public release. They define the correct confidence level for it.

11 Future Work

The next steps are straightforward.

- Independent validation on external corpora.
- Larger benchmark sets with more diverse question types.
- Direct comparison against classical RAG and common retrieval frameworks.
- Multi-model testing across GPT, Claude, GLM, Qwen, and DeepSeek.
- Publication of a benchmark summary CSV for easier reuse and inspection.

12 Conclusion

DCFMS Gateway shows credible potential as a context-reduction layer for LLM workflows. The public release package supports the idea that token usage, cost, and data exposure can be reduced while answer quality remains stable or improves after corpus refinement.

The goal is not only token reduction. The key objective is preserving answer quality while reducing context size. That is the right way to read the evidence in this release: the system is valuable when it reduces what reaches the model without sacrificing the usefulness of the answer.

At the same time, the evidence is still internal and requires further validation before any stronger scientific or operational claim should be made.

13 References / Source Materials

The following internal materials were used as source material for this publication snapshot.

Source material	Role in this paper
`docs/public_release_v0_2/README_PUBLIC.md`	Public-facing overview and release framing.
`docs/public_release_v0_2/EXECUTIVE_ONE_PAGER.md`	Short business summary and quality / cost framing.
`docs/public_release_v0_2/WHITEPAPER_v1.md`	Earlier whitepaper draft and narrative baseline.
`docs/public_release_v0_2/DCFMS_GATEWAY_PUBLIC_REPORT_v1.md`	Benchmark summary and evidence organization.
`docs/public_release_v0_2/PUBLIC_RELEASE_AUDIT.md`	Publication readiness, risk framing, and claims classification.
`docs/public_release_v0_2/PUBLIC_RELEASE_v0.3_CHANGELOG.md`	Privacy and data exposure wording alignment.
`benchmark_results/project_native_openai_full20_off_on.md`	Full20 OFF/ON quality baseline.
`benchmark_results/project_native_local_roi_token_compression.md`	Token-compression and ROI inputs.
`benchmark_results/client_demo_v1/FIRST_CLIENT_BENCHMARK.md`	Initial client-demo quality and compression results.
`benchmark_results/client_demo_v1/API_ASK_SMOKE.md`	Smoke benchmark after corpus patch.
`benchmark_results/client_demo_v1/API_ASK_FULL20_AFTER_PATCH.md`	Full20 after-patch client-demo benchmark.
`benchmark_results/client_demo_v1/CLIENT_DEMO_V1_FINAL_PATCH.md`	Final patch plateau evidence.
`benchmark_results/zero_drift_quality_ordered.md`	Laboratory ordered-context benchmark.
`docs/public_release_v0_2/figures/*.png`	Public figures used in the document.